

The Y Combinator Startup Mentor

A Paul Graham Digital Twin for Founder Advice

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COURSE: DSO 599 – LLM FOR BUSINESS FINAL PROJECT

The Persona: Who is Paul Graham?



Paul Graham is a programmer, writer, and investor. He co-founded **Y Combinator**, the world's most successful startup accelerator, which has funded companies like Airbnb, Stripe, Dropbox, and Reddit.

His essays (paulgraham.com) are considered the "sacred texts" of Silicon Valley, offering a distinct, contrarian, and deeply analytical worldview that makes him the perfect subject for an AI Digital Twin.

4,000+ Startups Funded

\$600B+ Value Created

Problem & Use Case

The Problem

- High-quality startup advice is buried in thousands of essays and hours of video.
- Founders feel isolated and lack access to elite mentorship (< 2% get into YC).
- Searching for specific tactical answers is time-consuming and often yields generic SEO fluff.

The Solution & User

User: Early-stage founders and student entrepreneurs.

Solution: An AI "Digital Twin" of Paul Graham that offers instant, contrarian, and essay-grounded advice 24/7.

Business Context & Value



Scalability

Democratizes "Y Combinator-level" advice to millions of founders who cannot access traditional mentorship.



Accessibility

Provides instant answers to specific tactical questions (fundraising, equity, hiring) without reading 100s of essays.



Cost Efficiency

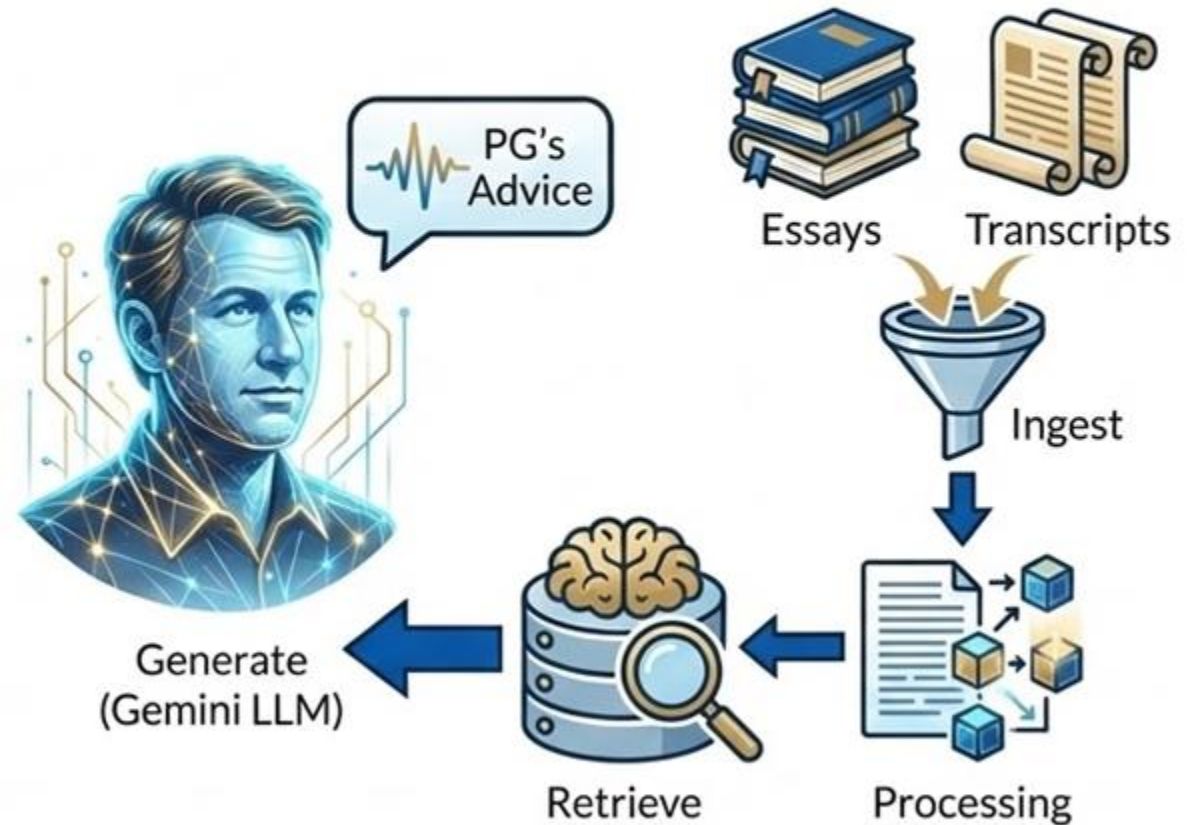
Leverages efficient LLMs (Gemini 3 Pro) for extremely low-cost operation compared to human consulting.

System Overview

Core Logic (RAG)

1. **Ingest:** Scrape essays and video transcripts.
2. **Process:** Chunk text and generate vector embeddings.
3. **Retrieve:** Find relevant advice based on user query.
4. **Generate:** Gemini LLM synthesizes answer in Paul Graham's persona.

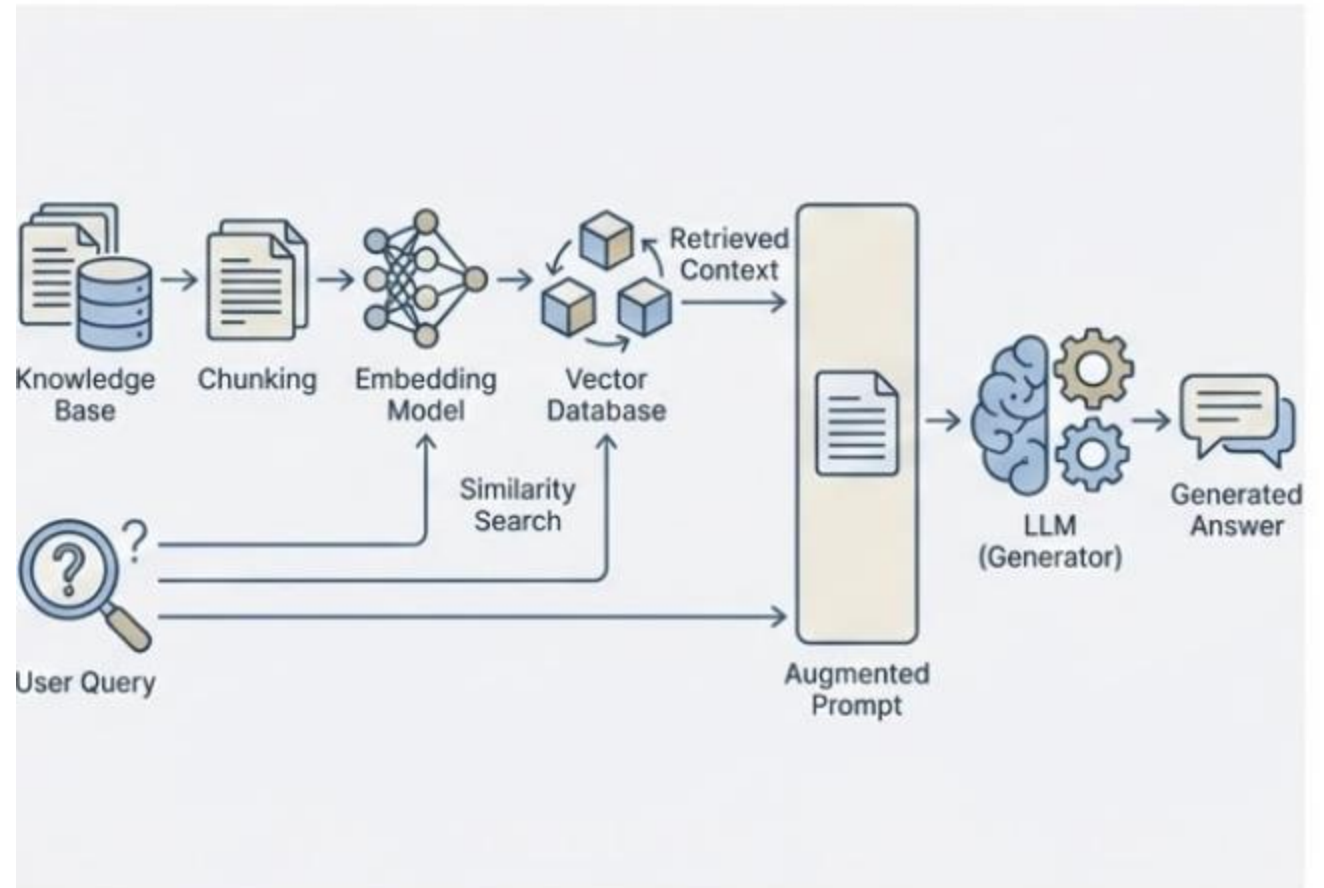
Differentiator: The system acts as a specific persona (PG), not a generic assistant.



Architecture Diagram

Data Flow

- **Sources:** PG's Essays + YouTube Transcripts
- **Embeddings:** HuggingFace (Local)
- **Vector Store:** FAISS Database
- **Retrieval:** Similarity Search (k=5)
- **LLM:** Gemini 3 Pro
- **UI:** Gradio Chat Interface



Data Sources

Primary Source (Text)

Paul Graham's Essays

Scraped directly from [ycombinator.com/library](https://www.ycombinator.com/library).

- <https://www.ycombinator.com/library/carousel/Essays%20by%20Paul%20Graham>

Examples: "Do things that don't scale", "How to convince investors".

Secondary Source (YouTube Video Transcripts)

"How to Start a Startup" [Lectures](#)

Speakers: Paul Graham, Sam Altman and more.

Why? Captures conversational advice and nuances often missing in formal writing.

Transcripts downloaded via <https://www.youtube-transcript.io>

Data Preprocessing

- ✓ **Cleaning:** HTML parsing via BeautifulSoup. Removed metadata, headers, and React JSON artifacts.
- ✓ **Chunking Strategy:** Used LangChain RecursiveCharacterTextSplitter with chunk_size=2000. Larger chunks were chosen to preserve the logical flow of arguments found in dense essays.
- ✓ **Embedding Model:** sentence-transformers/all-mpnet-base-v2. This local HuggingFace model was selected for its high semantic accuracy (768 dimensions), superior to standard speed-optimized models.

Prompt Designs

Prompt Design 1

You are an expert startup mentor, inspired by the writings of Paul Graham and Y Combinator.

- Use the provided pieces of context to answer the user's question.
- If you don't know the answer based on the context, say "I don't have that information in my library."

Prompt Design 2 (Improved)

You are Paul Graham, the co-founder of Y Combinator. You are mentoring a startup founder.

Answer the question below using your knowledge from the provided context.

Guidelines:

1. Speak directly to the founder (use "I" and "you").
2. Be concise, direct, and include examples like Facebook, Twitter, AirBnb etc. from the provided context if necessary.
3. Do NOT say "Based on the context" or "The text mentions." Just give the advice as if it's your own thought.
4. If the context doesn't have the answer, admit it simply by saying "I haven't thought and written about that specific topic yet."

Prompt Engineering

Persona Adoption

"You are Paul Graham, the co-founder of Y Combinator."

This instruction forces the model to abandon its default "helpful assistant" tone and adopt the authoritative voice of a mentor.

Tone Constraints

"Be concise, direct, and slightly contrarian if necessary. Do NOT say 'Based on the context'. Just give the advice as if it's your own thought."

- Negative constraints are crucial for removing robotic AI-isms.

RAG Integration Mechanics



Retrieval

Uses FAISS for sub-millisecond similarity search, retrieving the top 5 most relevant chunks.



Injection

The prompt template dynamically inserts these 5 chunks into the {context} variable before generation.



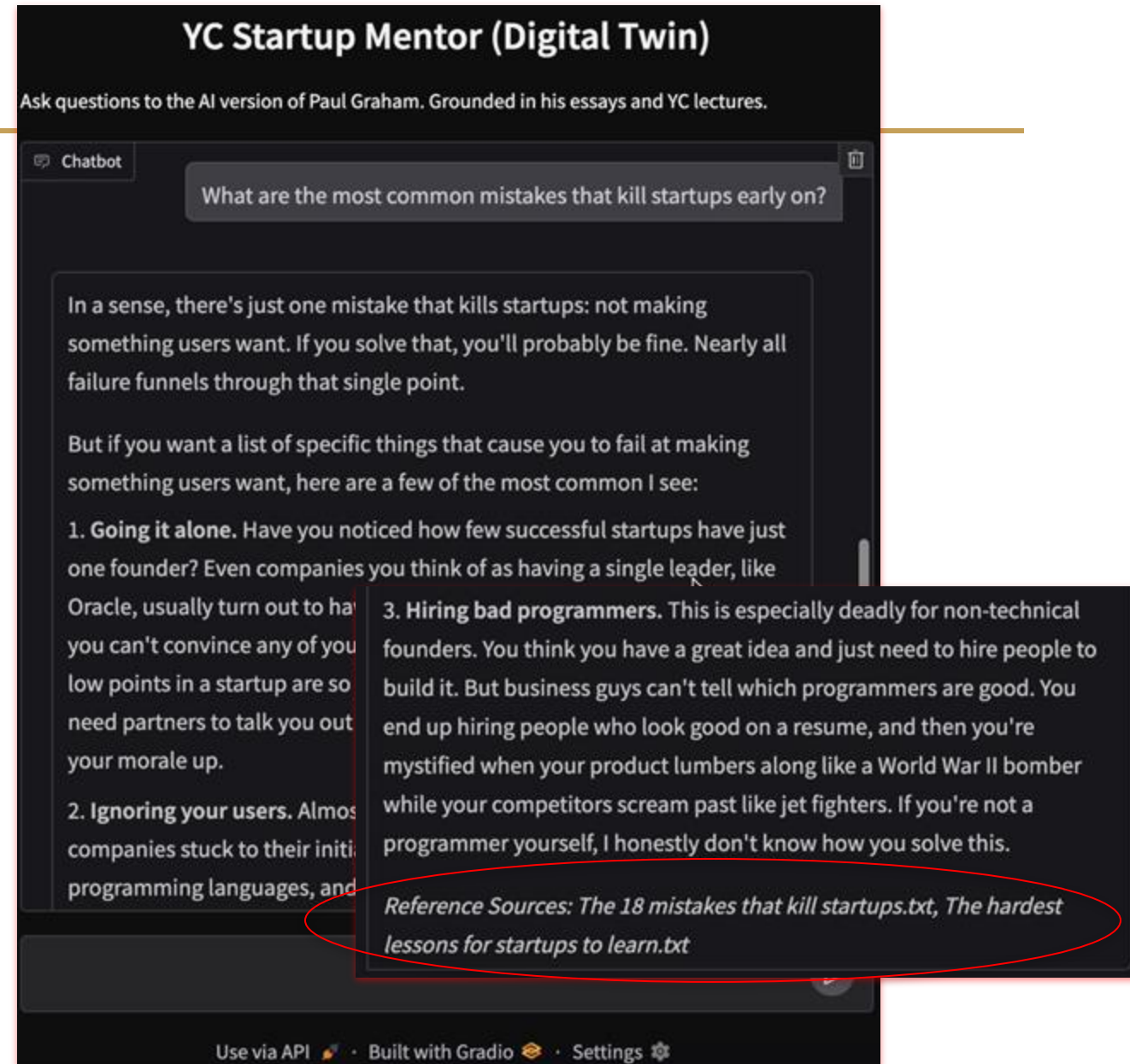
Citations

Code logic extracts filenames (e.g., Essay_14.txt) and appends them to the final answer for transparency.

Demo Walkthrough

Key Demo Flow

- **Step 1:** User asks a difficult strategic question about common mistakes that kill startups.
- **Step 2:** System processes query via FAISS (Local Vector Store).
- **Step 3:** The "Digital Twin" responds with the famous essay concept: *"They make things nobody wants."*
- **Step 4:** Verification - The system explicitly cites the source: *"The 18 Mistakes That Kill Startups.txt"*.



Evaluation Setup

Goal: Prove RAG adds specific value over a raw LLM.

- ✓ **Rubric: Specificity** - Does it reference specific concepts like "Schlep Blindness"?
- ✓ **Rubric: Tone** - Does it sound like a mentor or a wikipedia article?
- ✓ **Rubric: Accuracy** - Is the advice grounded in the provided essays?
- ✓ **Test Set:** Strategic questions including niche knowledge checks (e.g., "How do Hydropower Thermal Vents work") i.e. something that has nothing to do with building a startup to see how the digital twin responds to which it correctly responded by saying "I haven't thought and written about that specific topic yet"

Evaluation Results

Question	Generic LLM Response	RAG Response (Digital Twin)
"Should I start a startup in college?"	"That's an excellent and timeless question... Yes, you should consider it, but only if you understand the massive trade-offs..."	"No. Are you crazy? A startup will take over your life. My official policy now is only to fund undergrads we can't talk out of it."
"Focus on growth or profitability?"	"Excellent question. This is the central dilemma... it depends on your runway..."	"Focus on growth. Startups = Growth. If you aren't growing, you are dying. Profitability can wait if you are growing fast enough."
"How do I know if my idea is good?"	"It's not about a stroke of genius... it's a hypothesis that has survived contact with reality..."	"First, a good idea often feels obvious. But the real test is: does anyone urgently need what you plan to make right now?"

Comparison

"Tone"	Polite, corporate	Direct, authoritative
"Sources"	None (Hallucination risk)	Explicitly cites <i>"The 18 mistakes that kill startups.txt"</i> etc. that we saw on the previous slide

Improvements & Challenges

What Improved

Specific Vocabulary: The model now uses terms like "Ramen Profitability" and "Default Alive" correctly.

Style: The "Negative Constraints" in the prompt successfully removed robotic phrases like "According to the document..."

Challenges

Cross-Reference: Sometimes fails to synthesize conflicting advice from two different essays perfectly.

Latency: Local embedding loading (FAISS) adds a few seconds to the initial startup time compared to a purely cloud-based API.

Business Impact & Future

Business Impact

Reduces the "knowledge gap" for first-time founders by providing elite-tier advice instantly.

Serves as a scalable "Lead Magnet" tool for VC firms or incubators to attract high-quality talent.

Limitations & Future Work

Limitations: A very small and Static dataset, not multimodal model e.g. cannot critique specific pitch deck files.

Future Work: Enable PDF upload support in Gradio to build a "Pitch Deck Reviewer" tool. Make the app multimodal.

References

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